

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of M. Sc. (AOC) Examination September 2018

OC 729 Fundamentals of Physical Chemistry-II

Date: 19/09/2018 Day: Wednesday Time: 10.00 am To 10.30 am Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions.

[20]

1 At moderately low pressure, the fugacity of real gas is given by

(a) $f = \frac{PV}{RT}$ (b) $f = \frac{P^2V}{RT}$ (c) $f = \frac{PV^2}{RT}$

☐

2. The property of liquid arises due to resistance against the flow direction is known as:

- (a) Surface tension (b) Viscosity (c) Compressibility (d) Freezing

☐

3. Degree of polymerization (ρ) defined mathematically as: where M is the molecular weight of the polymer and m is the mass of a monomer.

- (a) $\rho = M + m$ (b) $\rho = M - m$ (c) $\rho = M / m$ (d) $\rho = M \times m$

☐

4. For a monoclinic crystal system, which is incorrect?

- (a) $a \neq b \neq c$ (b) $a = b = c =$ (c) $a = b \neq c$ (d) $a \neq b = c$

☐

5. The packing fraction for SC is

- (a) 0.74 (b) 0.68 (c) 0.52 (d) 0.30

☐

6. An Extensive property is depend upon

- (a) nature of the substance (b) quantity of matter
(c) external temperature (d) atmospheric pressure

☐

7. If there are 2 atoms in unit cell in a cubic system, it is an example of
 (a) simple cubic unit cell (b) body centred cubic unit cell
 (c) face centred cubic unit cell (d) none of these ☐
8. The energy associated with a photon is given by the equation
 (a) $E = h \times \lambda$ (b) $E = h \times \nu$ (c) $E = h \times c$ (d) $E = h \times c^2$ ☐
9. In Bragg's equation $n \lambda = 2 d \sin \theta$, ' θ ' represents
 (a) the number of moles (b) the principal quantum number
 (c) the incident angle (d) the order of reflection ☐
10. The emission of light as a result of chemical action is called _____
 (a) Phosphorescence (b) Fluorescence (c) Chemiluminescence (d) none of these ☐
11. The value R in joule is _____
 (a) 1.987 (b) 8.314 (c) 0.082 (d) 4.184 ☐
12. The SI unit of surface tension is _____
 (a) N/m (b) N.m (c) m/N (d) N ☐
13. Bragg's law related with:
 (a) Reflection (b) Refraction (c) Diffraction (d) Interference ☐
14. Which is natural polymer?
 (a) PVC (b) PS (c) Nylon (d) Teflon ☐
15. The number of molecules reacted or formed per photon of light absorbed is called
 (a) yield of the reaction (b) quantum efficiency (c) quantum yield (d) quantum productivity ☐
16. If intrinsic viscosity is $100 \text{ cm}^3/\text{g}$ and reduced viscosity is 0.5 then concentration is _____
 (a) 0.002 g/ml (b) 0.0025 g/ml (c) 0.005 g/ml (d) 0.5 ☐
17. A liquid rises in a capillary tube. It is due to
 (a) surface tension (b) viscosity (c) osmosis (d) effusion ☐
18. Number of atom per unit cell of SC crystal is:
 (a) 2 (b) 1 (c) 6 (d) 8 ☐
19. In Isotactic Polymer the arrangement of groups in chain are
 (a) Same side (b) Random Position (c) Alternate side (d) cross linked ☐
20. _____ stops as soon as the incident radiation is cut off
 (a) fluorescence (b) phosphorescence (c) chemiluminescence (d) none of these ☐

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OC 729 Fundamentals of Physical Chemistry-II

Date: 19/09/2017 Day: Wednesday Time: 10.30 am To 01.00 pm

Maximum Marks: 50

Instructions to the candidates:

1. **Section-I** and **Section-II** must be attempted in **Two answer sheet**.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of **non-programmable calculator** is **allowed**.
4. Show necessary calculation.

SECTION-I

[20]

Q-II

[A] Derive the equation for graphical method for determination of fugacity of gas. [05]

OR

[A] Derive the equation for approximate method. For a gas at 0°C $PV = RT (1 - 9.93 \times 10^{-4} P + 2.19 \times 10^{-6} P^2)$ calculate the fugacity at 50 atm. pressure [05]

[B] What if fugacity? Derive the equation $dF = RTd\ln f$ and equation for relative fugacity [05]

Q-III

[A] Discuss the mechanism of free radical Polymerization. [06]

[B] Discuss the lowering of freezing point method for determination mol. wt. of polymer. [04]

OR

[B] Write the classification of polymer in detail. [04]

SECTION-II

[30]

Q-IV

[A] Discuss the Jabblonski Diagram for various photo physical process. [05]

[B] What is Chemiluminescence? Discuss its mechanism with suitable example. [05]

OR

[B] Discuss the kinetics of photochemical reaction of $H_2 + Br_2 \rightarrow 2HBr$ [05]

Q-V

[A] Explain the drop weight and drop number method for determination of surface tension. [05]

OR

[A] How we can predict molecular structure with help of dipole moment. Explain with suitable examples. [05]

[B] Discuss the method for determination of viscosity. Explain the effect of temperature on viscosity. [05]

Q-VI

[A] What is diffraction? Derive the equation for Bragg's law. [05]

[B] What is plane of symmetry and axis of symmetry? Give the difference between amorphous solid and crystalline solid. [05]

OR

[B] Calculate the packing fraction and coordination number for Body centered cubic and Face centered cubic. [05]

Useful Physical Parameters:

Planck's constant, $h = 6.626068 \times 10^{-34}$ J.s

Universal gas constant, $= 8.31$ J/mol. K

Avogadro number, $N_A = 6.023 \times 10^{23}$

Boltzmann Constant, $k = 1.38 \times 10^{-23}$ J/K

Speed of light in vacuum, $c = 3.0 \times 10^8$ m/s